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Mobile bracket

Abstract

The present disclosure provides a mobile bracket including a bottom base, a first supporting rod, a second supporting rod, a pitch adjustment assembly and a rotation adjustment assembly. The bottom base is connected to a bottom end of the first supporting rod. The second supporting rod is connected to a top end of the first supporting rod. The pitch adjustment assembly is connected to one end of the second supporting rod far from the first supporting rod. The pitch adjustment assembly is capable of rotating relative to the second supporting rod to adjust a pitch angle. The pitch adjustment assembly includes a mounting plate. The rotation adjustment assembly is rotatably connected to the mounting plate. The rotation adjustment assembly is capable of rotating around an axis perpendicular to the mounting plate to adjust a rotation angle. The rotation adjustment assembly is configured to be connected to a display.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority under 35 U.S.C. § 119(a) to Chinese Patent Application No. 202321860846.3, filed Jul. 14, 2023, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a technical field of display mounting, and in particular relates to a mobile bracket.

BACKGROUND

(3) Existing floor stands for mounting a display can realize a height lifting adjustment and a left and right rotating adjustment. But they can not realize angle adjustments in other degrees of freedom. Therefore, existing floor stands cannot adapt to a variety of occasions of adjustments for display angle.

SUMMARY

(4) For this purpose, the present disclosure proposes a mobile bracket to solve at least one of the above mentioned technical problems.

(5) In order to solve the above mentioned technical problems, the technical solutions of the present disclosure are as follows.

(6) The present disclosure provides a mobile bracket. The mobile bracket includes a bottom case, a first supporting rod, a second supporting rod, a pitch adjustment assembly, and a rotation adjustment assembly. The bottom case is connected to a bottom end of the first supporting rod. The second supporting rod is connected to a top end of the first supporting rod. The pitch adjustment assembly is connected to one end of the second supporting rod far from the first supporting rod. The pitch adjustment assembly is capable of rotating up and down relative to the second supporting rod so as to adjust a pitch angle. The pitch adjustment assembly includes a mounting plate located at one end far from the first supporting rod. The rotation adjustment assembly is rotatably connected to the mounting plate. The rotation adjustment assembly is capable of rotating around an axis perpendicular to the mounting plate so as to adjust a rotation angle. The rotation adjustment assembly is configured to be connected to a back of a display.

(7) The beneficial effect of the present disclosure compared to the prior art is as follows.

(8) In the present disclosure, the pitch adjustment assembly which is capable of rotating up and down relative to the second supporting rod is adopted to adjust a pitch angle; the rotation adjustment assembly which is capable of rotating around an axis perpendicular to the mounting plate is adopted to adjust a rotation angle. Therefore, the present disclosure not only can adjust the pitch angle of the display, but also can adjust the rotation angle of the display to realize horizontal and vertical screen conversion, thus increasing the degrees of freedom in the angle adjustment of the display.

Description

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

(1) To describe the technology solutions in the embodiments of the present disclosure more clearly, the following briefly introduces the accompanying drawings required for describing the embodiments. Obviously, the accompanying drawings in the following description show merely at least one embodiment of the present disclosure, those of ordinary skilled in the art may also obtain other drawings based on these drawings without any creative efforts.

(2) FIG. 1 illustrates a three-dimensional structural schematic diagram of a mobile bracket in accordance with a first embodiment of the present disclosure;

(3) FIG. 2 illustrates an exploded diagram of the mobile bracket in FIG. 1;

(4) FIGS. 3A and 3B illustrate schematic diagrams of pitch angle adjustment of the mobile bracket in accordance with the first embodiment of the present disclosure;

(5) FIG. 4 illustrates a cross-sectional three-dimensional schematic diagram of FIG. 1 taken along a line IV-IV;

(6) FIG. 5 illustrates a partially enlarged view of FIG. 4 at V;

(7) FIG. 6 illustrates an exploded diagram of a pitch adjustment assembly in accordance with the first embodiment of the present disclosure;

(8) FIG. 7 illustrates a cross-sectional three-dimensional schematic diagram of FIG. 1 taken along a line VII-VII;

(9) FIGS. **8A** and **8B** illustrate schematic diagrams of rotation angle adjustment of the mobile bracket in accordance with the first embodiment of the present disclosure;

(10) FIG. **9** illustrates a schematic diagram of threading of the mobile bracket in accordance with the first embodiment of the present disclosure;

(11) FIG. **10** illustrates a structural schematic diagram of a bottom base of the mobile bracket in accordance with the first embodiment of the present disclosure;

(12) FIG. **11** illustrates a three-dimensional structural schematic diagram of the mobile bracket in accordance with a second embodiment of the present disclosure;

(13) FIG. **12** illustrates an exploded diagram of FIG. **11**;

(14) FIG. **13** illustrates a cross-sectional three-dimensional schematic diagram of the mobile bracket in accordance with the second embodiment of the present disclosure;

(15) FIG. **14** illustrates a schematic diagram of height adjustment of the mobile bracket in accordance with the second embodiment of the present disclosure;

(16) FIG. **15** illustrates a schematic diagram of threading of the mobile bracket in accordance with the second embodiment of the present disclosure;

(17) FIG. **16** illustrates a schematic diagram of pitch angle adjustment of the mobile bracket in accordance with the second embodiment of the present disclosure;

(18) FIG. **17** illustrates a schematic diagram of rotation angle adjustment of the mobile bracket in accordance with the second embodiment of the present disclosure;

(19) FIG. **18** illustrates a structural schematic diagram of a bottom base of the mobile bracket in accordance with the second embodiment of the present disclosure;

(20) FIG. **19** illustrates a three-dimensional structural schematic diagram of the mobile bracket in accordance with a third embodiment of the present disclosure;

(21) FIG. **20** illustrates an exploded diagram of FIG. **19**;

(22) FIG. **21** illustrates a cross-sectional three-dimensional schematic diagram of the mobile bracket in accordance with a third embodiment of the present disclosure;

(23) FIG. **22** illustrates a schematic diagram of height adjustment of the mobile bracket in accordance with the third embodiment of the present disclosure;

(24) FIG. **23** illustrates a schematic diagram of threading of the mobile bracket in accordance with the third embodiment of the present disclosure;

(25) FIG. **24** illustrates a schematic diagram of pitch angle adjustment of the mobile bracket in accordance with the third embodiment of the present disclosure;

(26) FIG. **25** illustrates a schematic diagram of rotation angle adjustment of the mobile bracket in accordance with the third embodiment of the present disclosure;

(27) FIG. **26** illustrates a schematic diagram of a bottom case of the mobile bracket in accordance with the third embodiment of the present disclosure;

(28) FIG. **27** illustrates a three-dimensional structural schematic diagram of the mobile bracket from another view in accordance with the third embodiment of the present disclosure;

(29) FIG. **28** illustrates a three-dimensional structural schematic diagram of the mobile bracket from one view in accordance with a fourth embodiment of the present disclosure;

(30) FIG. **29** illustrates an exploded diagram of FIG. **28**;

(31) FIG. **30** illustrates a three-dimensional structural schematic diagram of the mobile bracket from another view in accordance with the fourth embodiment of the present disclosure;

(32) FIG. **31** illustrates an exploded diagram of FIG. **30**;

(33) FIG. **32** illustrates a schematic diagram of an internal structure of a lifting arm after partial shell is removed in accordance with a fourth embodiment of the present disclosure;

(34) FIG. **33** illustrates a cross-sectional three-dimensional schematic diagram of FIG. **30** at XXXIII;

(35) FIG. **34** illustrates a partially enlarged view of FIG. **33** at XXXIV;

(36) FIG. **35** illustrates a schematic diagram of height adjustment of the mobile bracket in

accordance with a fourth embodiment of the present disclosure;

(37) FIG. **36** illustrates a schematic diagram of pitch angle adjustment of the mobile bracket in accordance with the fourth embodiment of the present disclosure;

(38) FIG. **37** illustrates a schematic diagram of rotation angle adjustment of the mobile bracket in accordance with the fourth embodiment of the present disclosure;

(39) FIG. **38** illustrates a schematic diagram of a bottom base of the mobile bracket in accordance with the fourth embodiment of the present disclosure;

(40) FIG. **39** illustrates a schematic diagram of threading of the mobile bracket in accordance with the fourth embodiment of the present disclosure.

DETAILED DESCRIPTION OF ILLUSTRATED EMBODIMENTS

(41) Embodiments of the present disclosure are described in detail below. Examples of the embodiments are shown in the accompanying drawings. The same or similar labeling throughout denotes the same or similar elements or elements having the same or similar functions. The embodiments described below by reference to the accompanying drawings are exemplary and are intended to be configured to explain the present disclosure and are not to be construed as limiting the present disclosure.

(42) Furthermore, the terms “first” and “second” are used for descriptive purposes only and are not to be understood as indicating or implying relative importance or implicitly specifying the number of technical features indicated. Therefore, a feature defined with “first” or “second” may include one or more such features, either explicitly or implicitly. In the description of the present disclosure, “more than one” means two or more, unless otherwise expressly and specifically limited.

(43) In the present disclosure, unless otherwise expressly specified and limited, the terms “connected”, “fixed”, and the like are to be understood broadly, e.g., they may be fixedly connected, removably connected, or integrally connected: they may be mechanically connected or electrically connected: they may be directly connected or indirectly connected through an intermediate medium; they may be interconnected within two elements or an interactive relationship between two elements. For those of ordinary skill in the art, the specific meaning of the above terms in the present disclosure may be understood on a case-by-case basis.

(44) Referring to FIGS. **1** and **2**, FIG. **1** illustrates a three-dimensional structural schematic diagram of a mobile bracket in accordance with a first embodiment of the present disclosure; and FIG. **2** illustrates an exploded diagram of the mobile bracket in FIG. **1**.

(45) For convenience of description, a height direction of the mobile bracket **1** shown in FIG. **1** is defined as a Z-axis direction, with a positive direction toward the Z-axis direction as a “top”, and a negative direction toward the Z-axis direction as a “bottom”, which does not constitute a limitation of the mobile bracket **1** in practical application scenarios.

(46) As shown in FIGS. **1** and **2**, the mobile bracket **1** includes a bottom base **11**, a first supporting rod **12**, a second supporting rod **13**, a pitch adjustment assembly **14** and a rotation adjustment assembly **15**. The bottom base **11** is connected to a bottom end of the first connecting rod **12**. The second supporting rod **13** is connected to a top end of the first supporting rod **12**. The pitch adjustment assembly **14** is connected to one end of the second supporting rod **13** far from the first supporting rod **12**. The pitch adjustment assembly **14** is capable of rotating up and down relative to the second supporting rod **13** so as to adjust a pitch angle. The pitch adjustment assembly **14** includes a mounting plate **141** located at one end far from the first supporting rod **12**. The rotation adjustment assembly **15** is rotatably connected to the mounting plate **141**. The rotation adjustment assembly **15** is capable of rotating around an axis perpendicular to the mounting plate **141** so as to adjust a rotation angle. The rotation adjustment assembly **15** is configured to be connected to a back of a display (not shown).

(47) In the present disclosure, the pitch adjustment assembly **14** which is capable of rotating up and down relative to the second supporting rod **13** is adopted to adjust a pitch angle: the rotation

adjustment assembly **15** which is capable of rotating around an axis perpendicular to the mounting plate **141** is adopted to adjust a rotation angle. Therefore, the present disclosure not only can adjust the pitch angle of the display, but also can adjust the rotation angle of the display to realize horizontal and vertical screen conversion, thus increasing the degrees of freedom in the angle adjustment of the display.

(48) In at least one embodiment, referring to FIGS. **3A** and **3B**, the range of the pitch adjustment angle of the pitch adjustment assembly **14** is -20° to 30° . The angle is an angle between the mounting plate **141** and a center axis of the first supporting rod **12**. It can be understood that in other embodiments, a range of the pitch adjustment angle may be adjusted according to an actual need and is not limited herein.

(49) Thereby, when the pitch adjustment angle of the pitch adjustment assembly **14** is within a range of -20° to 30° , the pitch adjustment needs in more application scenarios can be met.

(50) In at least one embodiment, referring to FIGS. **2**, **4** and **5**, the pitch adjustment assembly **14** includes a rotating shaft **142** and a connecting arm **140**. The rotating shaft **142** is connected to one end of the second supporting rod **13** far from the first supporting rod **12**. The connecting arm **140** is rotatably connected to the rotating shaft **142**. The connecting arm **140** is capable of rotating around a center axis of the rotating shaft **142**, thereby achieving a need for adjusting pitch angle.

(51) In at least one embodiment, the connecting arm **140** is substantially U-shaped. The connecting arm **140** includes the mounting plate **141**, a first arm **143** and a second arm **144** extending coextensively from a surface of the mounting plate **141**. The first arm **143** and the second arm **144** are spaced apart from each other. The first arm **143** defines a first through hole **1431**. The second arm **144** defines a second through hole **1441**. The second supporting rod **13** defines a third through hole **131** and a fourth through hole **132**. The rotating shaft **142** passes through the second through hole **1441**, the third through hole **131**, the first through hole **1431**, and the fourth through hole **132** to rotatably connect the first arm **143** and the second arm **144** to the end of the second supporting rod **13** far from the first supporting rod **12**.

(52) Thereby, the rotating shaft **142** can rotatably connect the connecting arm **140** to the second supporting rod **13** at the end of the second supporting rod **13** far from the first supporting rod **12**, achieving to adjust the pitch angle of the display.

(53) In at least one embodiment, referring to FIGS. **2** and **5** together, the pitch adjustment assembly **14** further includes a torsion spring **145** and a limiting rod **1460**. The torsion spring **145** is threaded onto the rotating shaft **142**. The torsion spring **145** includes a first free end **1451** and a second free end **1452** extending in opposite directions. The first free end **1451** resists against a top wall within the second supporting rod **13**. The limiting rod **1460** is secured between the first arm **143** and the second arm **144**, and is located between the rotating shaft **142** and the mounting plate **141** and is parallel to the rotating shaft **142**. The second free end **1452** is held against a lower side of the limiting rod **1460**.

(54) Therefore, when the rotation adjustment assembly **15** and the display as a whole are adjusted to an upward elevation state, as the rotation adjustment assembly **15** and the display have a certain weight, there will be a tendency for a downward movement relative to the second supporting rod **13**, the second free end **1452** of the torsion spring **145** resists against the lower side of the limiting rod **1460**, which can at least partially offset the gravity force and prevent the pitch angle of the display from changing due to gravity force when the display is adjusted to the upward elevation state.

(55) In at least one embodiment, referring to FIGS. **2**, **6** and **7** together, the pitch adjustment assembly **14** further includes a first shell **146** and a second shell **147**. The first shell **146** is connected to the first arm **143**. The second shell **147** is connected to the second arm **144**. The first shell **146** has a first hook **1461** protruding from an inner wall of the first shell **146**. The first arm **143** defines a first hole **1432** corresponding to a position of the first hook **1461**. When the first shell **146** is connected to an outside of the first arm **143**, the first hook **1461** passes through the first hole

1432 and snaps onto the inner wall of the first arm **143**. The second shell **147** has a second hook **1471** protruding from an inner wall of the second arm **147**. The second arm **144** defines a second hole **1442** corresponding to a position of the second hook **1471**. When the second shell **147** is connected to an outside of the second arm **144**, the second hook **1471** passes through the second hole **1442** and snaps onto the inner wall of the second arm **144**.

(56) Thereby, the first shell **146** can be snapped onto the first arm **143**, and the second shell **147** can be snapped onto the second arm **144**; the connecting arm **140** is covered by the first shell **146** and the second shell **147** for aesthetic purposes.

(57) In at least one embodiment, the first shell **146** includes a first snap portion **1462** at a position far from the mounting plate **141**. The first arm **143** includes a first inclined portion **1433** at a position corresponding to the first snap portion **1462**. The first snap portion **1462** is snapped onto the first inclined portion **1433**. The second shell **147** includes a second snap portion **1472** at a position far from the mounting plate **141**. The second arm **144** includes a second inclined portion **1443** at a position corresponding to the second snap portion **1472**. The second snap portion **1472** is snapped onto the second inclined portion **1443**.

(58) Thereby, the first shell **146** can be further secured by the first snap portion **1462** being snapped onto the first inclined portion **1433**, and the second shell **147** can be further secured by the second snap portion **1472** being snapped onto the second inclined portion **1443**, such that the first shell **146** and the second shell **147** can be mounted more stably.

(59) In at least one embodiment, referring to FIGS. **5** and **6**, portions of the first shell **146** and the second shell **147** corresponding to the second supporting rod **13** have hollowed spaces.

(60) When the pitch adjustment assembly **14** is rotated upwardly relative to the second supporting rod **13** to a certain angle, a travel of the pitch adjustment assembly **14** in its upward rotational movement will be limited by the first shell **146** and the second shell **147**, so that an upward angle range of the pitch adjustment assembly **14** can be adjusted by adjusting a size of the hollowed spaces of the first shell **146** and the second shell **147**.

(61) In at least one embodiment, referring again to FIG. **2**, the rotation adjustment assembly **15** includes a rotating plate **151** and a connecting member **152**. The rotating plate **151** is configured to be connected to the back of the display. The connecting member **152** connects the mounting plate **141** and the rotating plate **151**. The rotating plate **151** and the connecting member **152** are stop-rotationally connected to each other, and the connecting member **152** can rotate relative to the mounting plate **141**.

(62) In a specific embodiment, the connecting member **152** includes a head portion **1521** and a connecting rod **1522**. The connecting rod **1522** is in a shape of a semi-cylindrical cylinder. The head portion **1521** and the connecting rod **1522** are connected. The mounting plate **141** defines a first connecting hole **1411**. The first connecting hole **1411** is a circular hole. A center of the rotating plate **151** defines a second connecting hole **1511**. The second connecting hole **1511** is a semicircular hole. The head portion **1521** resists against one end of the mounting plate **141** far from the rotating plate **151**. The connecting rod **1522** passes through the first connecting hole **1411** as well as the second connecting hole **1511** and is then connected to a nut.

(63) Thereby, the rotating plate **151** and the connecting member **152** are stop-rotationally connected, the connecting member **152** can rotate relative to the mounting plate **141**, realizing a rotatable connection between the rotating plate **151** and the mounting plate **141**. The rotating plate **151** is capable of rotating relative to a center axis of the connecting member **152**.

(64) In at least one embodiment, the mounting plate **141** includes a protrusion **1412** on a side of the mounting plate **141** facing the rotating plate **151**. The rotating plate **151** defines a guiding groove **1512** corresponding to the protrusion **1412**. The guiding groove **1512** is a curved groove centered on a center of the rotating plate **151**. The protrusion **1412** is inserted into the guiding groove **1512** to form a limit to a rotational travel of the rotation adjustment assembly **15**.

(65) In a specific embodiment, referring to FIGS. **8A** and **8B** together, when the protrusion **1412**

does not move, rotating the rotating plate **151** can cause the guiding groove **1512** of the rotating plate **151** to exhibit a different positional relationship to the protrusion **1412**, i.e., such that the protrusion **1412** is located at a different position of the guiding groove **1512**. In FIG. **8A**, the rotating plate **151** has a rotation angle of 0° . In FIG. **8B**, the rotational disk **151** has a rotation angle of 90° . Thus, the rotation angle of the rotation adjustment assembly **15** ranges from 0° to 90° . Thereby, by adjusting the rotation angle of the rotary disk **151**, horizontal and vertical screen conversions of the display can be realized, so that the viewing angle of the display can be switched at will in different scenarios, so as to provide the user with a better using experience.

(66) In at least one embodiment, referring again to FIG. **7**, the rotating plate **151** includes a center zone **1513**, a first step portion **1514**, and a second step portion **1515**. The first step portion **1514** is located around the center zone **1513**. The second step portion **1515** is located around the first step portion **1514**. The rotation adjustment assembly **15** further includes a shield **153**. The shield **153** defines a through hole **1531**. A diameter of the through hole **1531** is greater than or equal to a diameter of the center zone **1513**. The through hole **1531** has an inner wall adjacent to an outer wall of the first step section **1514**. The shield **153** gradually slopes to a side proximate the rotating plate **151** from a center of the shield **153** to an edge of the shield **153**.

(67) In at least one embodiment, the shield **153** defines a plurality of heat dissipation holes **1532** for heat dissipation from peripheral elements.

(68) In at least one embodiment, the shield **153** further includes a bent convex edge **1533**. The bent convex edge **1533** is adjacent to a periphery of the rotating plate **151** and covers a periphery of the second step portion **1515** far from the first step portion **1514**.

(69) Therefore, the shield **153** can cover internal components fully, which can increase aesthetics.

(70) In at least one embodiment, referring to FIG. **9**, the mobile bracket **1** further includes a power supply assembly **16**. The power supply assembly **16** is located within the bottom base **11**. The power supply assembly **16** may be, but is not limited to, a safety circuit, a transformer circuit, etc. The first supporting rod **12** defines a first threading hole **121**. The second supporting rod **13** defines a second threading hole **130**. The pitch adjustment assembly **14** defines a third threading hole (not shown). The first threading hole **121**, the second threading hole **130** and the third threading hole are internally connected. A wire connected between the power supply assembly **16** and the display connected to the rotation adjustment assembly **15** passes through the first threading hole **121**, the second threading hole **130** and the third threading hole. The power supply assembly **16** is configured to be connected to a municipal circuit to supply power to the display which is mounted to the mobile bracket **1**.

(71) Therefore, through the first threading hole **121**, the second threading hole **130** and the third threading hole, an internal wiring can be realized. The power supply assembly **16** for powering the display is fixed to the bottom base **11**. connecting the display through internal wiring can make the mobile bracket **1** appearance more beautiful and simple.

(72) In at least one embodiment, referring to FIG. **10**, the mobile bracket **1** further includes a fixing member **111**. The fixing member **111** includes a supporting body **1111**, a first folding portion **1112** and a second folding portion **1113**. The first folding portion **1112** and the second folding portion **1113** are connected to opposite sides of the supporting body **1111**. The first folding portion **1112** and the second folding portion **1113** are L-shaped respectively and secured to a lower surface of the bottom base **11**. A space between the supporting body **1111**, the first folding portion **1112** and the second folding portion **1113** is configured for mounting the power supply assembly **16**.

(73) In at least one embodiment, the fixing member **111** further includes a first stopping portion **1114** and a second stopping portion **1115** from opposite additional sides of the supporting body **1111**. The first folding portion **1112**, the second folding portion **1113**, the first stopping portion **1114**, and the second stopping portion **1115** respectively limit the power supply assembly **16** from four sides of the power supply assembly **16**.

(74) Therefore, the fixing member **111** can securely and reliably fix the power supply assembly **16**

to the bottom surface of the bottom base **11**. In at least one embodiment, there is a battery mounting position reserved in the bottom base **11**, allowing the display to be free from wiring harnesses and to be used anytime, anywhere, and at will.

(75) In at least one embodiment, the mobile bracket **1** further includes at least three roller sets **17**. The at least three roller sets **17** are located at different positions on the lower surface of the bottom base **11** and spaced apart from the power supply assembly **16**.

(76) Thereby, a rapid movement of the mobile bracket **1** can be realized by the at least three roller sets **17**.

(77) In at least one embodiment, the mobile bracket **1** further includes a shielding ring **112**. The shielding ring **112** is bent downwardly from a periphery of the bottom base **11**. A distance between a bottom end of the shielding ring **112** and a ground is smaller than a distance between the bottom base **11** and the ground.

(78) Therefore, the at least three roller sets **17** can be obscured by the shielding ring **112**, and the at least three roller sets **17** cannot be seen from the outside from a user's view, which can make the mobile bracket **1** more aesthetically pleasing and simple. The shielding ring **112** is almost near the ground, which can not only keep the mobile bracket **1** moving by using the at least three roller sets **17** but also ensure the aesthetics of the mobile bracket **1**.

(79) In at least one embodiment, a beveled edge design is configured between the bottom base **11** and the shielding ring **112**, which can make the product appearance more beautiful and neat.

(80) It can be understood that in the first embodiment, the height of the first supporting rod **12** remains constant, i.e., the height of the first supporting rod **12** cannot be adjusted.

(81) Referring together to FIGS. **11** to **18**, FIG. **11** illustrates a three-dimensional structural schematic diagram of the mobile bracket in accordance with a second embodiment of the present disclosure; FIG. **12** illustrates an exploded diagram of FIG. **11**; FIG. **13** illustrates a cross-sectional three-dimensional schematic diagram of the mobile bracket in accordance with the second embodiment of the present disclosure; FIG. **14** illustrates a schematic diagram of height adjustment of the mobile bracket in accordance with the second embodiment of the present disclosure; FIG. **15** illustrates a schematic diagram of threading of the mobile bracket in accordance with the second embodiment of the present disclosure; FIG. **16** illustrates a schematic diagram of pitch angle adjustment of the mobile bracket in accordance with the second embodiment of the present disclosure; FIG. **17** illustrates a schematic diagram of rotation angle adjustment of the mobile bracket in accordance with the second embodiment of the present disclosure; FIG. **18** illustrates a structural schematic diagram of a bottom base of the mobile bracket in accordance with the second embodiment of the present disclosure. The mobile bracket **1** in the second embodiment is structurally similar to the mobile bracket **1** in the first embodiment, with the difference that the height of the first supporting rod **12** of the mobile bracket **1** in the second embodiment is adjustable.

(82) As shown in FIGS. **11** and **12**, the first supporting rod **12** includes a supporting arm **122**, a lifting arm **123**, and a height adjustment member **124**. The lifting arm **123** is inserted at one end of the supporting arm **122** far from the bottom base **11**. The height adjusting member **124** is located within the supporting arm **122** and the lifting arm **123**. The height adjusting member **124** adjusts a protruding length of the lifting arm **123** relative to the supporting arm **122** to adjust a total height between the supporting arm **122** and the lifting arm **123**.

(83) Referring to FIG. **12**, in the second embodiment, the height adjusting member **124** is a damping gas spring, which can be stopped at any position in travel without any external structure. The damping gas spring is a constant force gas spring, i.e., it can only withstand a preset weight, for example, when the factory pressure of the damping gas spring is set to 5 KG, the mobile bracket **1** can only be fitted with a display having a weight of 5 KG. For another example, when the factory pressure of the damping gas spring is set to 3 KG, the mobile bracket **1** can only be fitted with a display having a weight of 3 KG.

(84) Thus, by using the damping gas spring, the lifting arm **123** can be fixed at any height at any time to meet the user's needs, so that the mobile bracket **1** can be adapted to a wider range of display sizes.

(85) Referring to FIGS. **19** to **27**, FIG. **19** illustrates a three-dimensional structural schematic diagram of the mobile bracket in accordance with a third embodiment of the present disclosure; FIG. **20** illustrates an exploded diagram of FIG. **19**; FIG. **21** illustrates a cross-sectional three-dimensional schematic diagram of the mobile bracket in accordance with a third embodiment of the present disclosure; FIG. **22** illustrates a schematic diagram of height adjustment of the mobile bracket in accordance with the third embodiment of the present disclosure; FIG. **23** illustrates a schematic diagram of threading of the mobile bracket in accordance with the third embodiment of the present disclosure; FIG. **24** illustrates a schematic diagram of pitch angle adjustment of the mobile bracket in accordance with the third embodiment of the present disclosure; FIG. **25** illustrates a schematic diagram of rotation angle adjustment of the mobile bracket in accordance with the third embodiment of the present disclosure; FIG. **26** illustrates a schematic diagram of a bottom case of the mobile bracket in accordance with the third embodiment of the present disclosure; FIG. **27** illustrates a three-dimensional structural schematic diagram of the mobile bracket from another view in accordance with the third embodiment of the present disclosure. The mobile bracket **1** in the third embodiment is structurally similar to the mobile bracket **1** in the first embodiment, with the difference that the height of the first supporting rod **12** of the mobile bracket **1** in the third embodiment is adjustable.

(86) As shown in FIG. **20**, the height adjusting member **124** is a self-locking gas spring. The self-locking gas spring can be stopped at any position in a travel with an aid of a release mechanism, and has a large locking force (which can be more than 10,000 N) after stopping. It can be understood that the self-locking gas spring can have different load-bearing capacities: in different application scenarios, the self-locking gas spring **1240** can be selected for different load-bearing capacities according to demands. Specifically, in the third embodiment, the height adjusting member **124** includes a self-locking gas spring **1240** and a control button **125**. The control button **125** is the release mechanism. The self-locking gas spring **1240** is located within the supporting arm **122** and the lifting arm **123**. The control button **125** is connected to one end of the lifting arm **123** far from the supporting arm **122** and pressed against the self-locking gas spring **1240**. When the control button **125** is pressed, the height adjustment function of the self-locking gas spring **1240** can be unlocked, thereby enabling adjustment of a height of the lifting arm **123** protruding from the supporting arm **122** by lifting the lifting arm **123** upwardly or pressing the lifting arm **123** downwardly to adjust a protruding height of the lifting arm **123** relative to the supporting arm **122**. When the control button **125** is released, the height adjustment function of the self-locking gas spring **1240** can be locked, at which time the height of the lifting arm **123** relative to the supporting arm **122** cannot be adjusted. Therefore, to adjust the height of the lifting arm **123** relative to the supporting arm **122**, it is necessary to keep pressing the control button **125** until the height of the lifting arm **123** relative to the supporting arm **122** is adjusted to a desired height, and then release the control button **125** to lock the height adjustment function of the self-locking gas spring **1240** again. The self-locking gas spring **1240** can be adapted to displays with different weights. For example, when the maximum load bearing capacity of the mobile bracket **1** is 15 KG, then the displays of any weight up to 15 KG are suitable for the mobile bracket **1**, and the height adjustment requires continuous pressing of the control button **125** to unlock the self-locking gas spring, before the height adjustment can be realized.

(87) Thereby, by controlling the lifting and lowering of the self-locking gas spring **1240** via the control button **125**, the total height between the supporting arm **122** and the lifting arm **123** can be adjusted at any time, so that the mobile bracket **1** can be adapted to fit a wider range of dimensions of the display.

(88) Due to the cooperation of the torsion spring **145** and the limiting rod **1460**, the mobile bracket

1 of the first embodiment, the second embodiment and the third embodiment can counteract a tendency to rotate downward due to the gravity of the display, and the adjustment of the angle of the upward tilt and the downward tilt is more labor-saving.

(89) Referring to FIGS. **28** to **39**, FIG. **28** illustrates a three-dimensional structural schematic diagram of the mobile bracket from one view in accordance with a fourth embodiment of the present disclosure; FIG. **29** illustrates an exploded diagram of FIG. **28**; FIG. **30** illustrates a three-dimensional structural schematic diagram of the mobile bracket from another view in accordance with the fourth embodiment of the present disclosure; FIG. **31** illustrates an exploded diagram of FIG. **30**; FIG. **32** illustrates a schematic diagram of an internal structure of a lifting arm after partial shell is removed in accordance with a fourth embodiment of the present disclosure; FIG. **33** illustrates a cross-sectional three-dimensional schematic diagram of FIG. **30** at XXXIII; FIG. **34** illustrates a partially enlarged view of FIG. **33** at XXXIV; FIG. **35** illustrates a schematic diagram of height adjustment of the mobile bracket in accordance with a fourth embodiment of the present disclosure; FIG. **36** illustrates a schematic diagram of pitch angle adjustment of the mobile bracket in accordance with the fourth embodiment of the present disclosure; FIG. **37** illustrates a schematic diagram of rotation angle adjustment of the mobile bracket in accordance with the fourth embodiment of the present disclosure; FIG. **38** illustrates a schematic diagram of a bottom base of the mobile bracket in accordance with the fourth embodiment of the present disclosure; FIG. **39** illustrates a schematic diagram of threading of the mobile bracket in accordance with the fourth embodiment of the present disclosure. The mobile bracket **1** in the fourth embodiment is structurally similar to the mobile bracket **1** in the first embodiment, with the difference that the height of the first supporting rod **12** of the mobile bracket **1** in the fourth embodiment can be adjusted, and the supporting arm **122** and the lifting arm **123** of the first supporting rod **12** can be tilted relative to each other, therefore, the total height of the supporting arm **122** and the lifting arm **123** can be adjusted by adjusting a tilt angle of the lifting arm **123** relative to the supporting arm **122**.

(90) Specifically, as shown in FIGS. **28** to **31**, the first supporting rod **12** includes a supporting arm **122**, a lifting arm **123**, and a tilt angle adjusting member **126**. The lifting arm **123** is inclined relative to the supporting arm **122**. The tilt angle adjusting member **126** is received in the lifting arm **123**. The tilt angle adjusting member **126** adjusts the tilt angle of the lifting arm **123** relative to the supporting arm **122**. The tilt angle adjusting member **126** causes the lifting arm **123** to be stayed at any angle within a predetermined range of angle adjustment and is capable of providing support for the suspension of the display.

(91) In at least one embodiment, referring to FIG. **32**, FIG. **33**, and FIG. **34** together, the tilt angle adjusting member **126** includes a first mounting base **1261**, a lower connecting rod **1262**, an upper tension spring **1263**, and a second mounting base **1264**. The first mounting base **1261** is connected to a top end of the supporting arm **122**, and the second mounting base **1264** is connected to one end of the second supporting rod **13** near the lifting arm **123**. In the fourth embodiment, the second mounting base **1264** and the second supporting rod **13** are integrally formed. The second supporting rod **13** extends upwardly at an angle from one end of the second mounting base **1264** far from the first mounting base **1261**. The lower connecting rod **1262** is rotatably connected to a bottom end of the first mounting base **1261** and a bottom end of the second mounting base **1264**. The upper tension spring **1263** is rotatably connected to a top end of the first mounting base **1261** and the bottom end of the second mounting base **1264**. The second mounting base **1264** is connected to the second supporting rod **13**. Specifically, the first mounting base **1261** defines a first mounting hole **1265** and a second mounting hole **1266**. The first mounting hole **1265** and the second mounting hole **1266** are positioned in an up-and-down relationship. The first mounting hole **1265** and the second mounting hole **1266** extend in a substantially perpendicular direction. The first mounting hole **1265** is located further far from the supporting arm **122** than the second mounting hole **1266**. The second mounting base **1264** defines a third mounting hole **1267** and a fourth

mounting hole **1268**. The third mounting hole **1267** and the fourth mounting hole **1268** are positioned in a generally left-right relationship. The fourth mounting hole **1268** is further far from the lifting arm **123** than the third mounting hole **1267**. The lower connecting rod **1262** is rotatably connected to the second mounting hole **1266** of the first mounting base **1261** and the third mounting hole **1267** of the second mounting base **1264**. The upper tension spring **1263** is rotatably connected to the first mounting hole **1265** of the first mounting base **1261** and the fourth mounting hole **1268** of the second mounting base **1264**.

(92) Therefore, in the fourth embodiment, the upper tension spring **1263** is configured inside the lifting arm **123**, and the upper tension spring **1263** plays a certain tensile force in the process of lifting or lowering of the lifting arm **123** so that the display can be stopped at any position.

(93) In at least one embodiment, referring to FIGS. **29** and **31**, the lifting arm **123** includes an upper shell **1231** and a bottom shell **1232**. The upper shell **1231** and the bottom shell **1232** are both semi-circular shells. It can be understood that in at least one embodiment, the upper shell **1231** and the bottom shell **1232** cooperate with each other to form the lifting arm **123**. It can be understood that in at least one embodiment, the tilt angle adjusting member **126** is connected to the supporting arm **122**, and the upper shell **1231** and the bottom shell **1232** of the lifting arm **123** are fixed to opposite sides of the tilt angle adjusting member **126**.

(94) Referring to FIG. **31**, the pitch adjustment assembly **14** includes a rotating shaft **142** and a connecting arm **140**. The rotating shaft **142** is fixed to the end of the second supporting rod **13** far from the first supporting rod **12**. The connecting arm **140** includes the mounting plate **141**, a first arm **143** and a second arm **144** formed by bending and extending in a same direction from opposite ends of the mounting plate **141**. The first arm **143** and the second arm **144** are spaced apart from each other. The rotating shaft **142** rotatably connects the first arm **143** and the second arm **144** to the end of the second supporting rod **13** far from the first supporting rod **12**. The mounting plate **141** is in a shape of a disk. The rotation adjustment assembly **15** includes a rotating plate **151** and a connecting member **152**. The connecting member **152** connects the mounting plate **141** and the rotating plate **151**. The rotating plate **151** is capable of rotating relative to the connecting member **152**. The rotating plate **151** is configured to be connected to the back of the display. Specifically, the first arm **143** defines a first through hole **1431**; and the second arm **144** defines a second through hole **1441**. The second supporting rod **13** is cylindrical in shape with a center axis of the cylinder parallel to a center axis of the rotating shaft **142**. The cylinder defines a third through hole and a fourth through hole **132** along opposite ends of its center axis. The third through hole is not labeled in FIGS. **29** and **31**. It can be understood that the third through hole and the fourth through hole **132** are symmetrically defined on opposite ends of the cylinder. The rotating shaft **142** includes two portions of rotating shaft, namely a first rotating shaft **1421** and a second rotating shaft **1422**. The first rotating shaft **1421** is received in the third through hole. The second rotating shaft **1422** is received in the fourth through hole **132**. The first rotating shaft **1421** and the second rotating shaft **1422** are butted together and stop-rotating, respectively, within the third through hole and the fourth through hole **132**, respectively.

(95) Referring to FIGS. **29** and **31**, one end of the connecting member **152** is connected to a cylindrical surface **133** of the cylinder and can swing up and down relative to a circumferential direction of the cylindrical surface **133**. Specifically, the cylindrical surface **133** of the cylinder defines a sliding groove **134**. The sliding groove **134** is defined extending along a circumferential direction of the cylindrical surface **133**. One end of the connecting member **152** is inserted into the sliding groove **134** and is guided by the sliding groove **134** to realize upward and downward sliding, and the other end of the connecting member **152** passes through the mounting plate **141** and is stop-rotationally connected to the mounting plate **141** and rotatably connected to the rotating plate **151**.

(96) Therefore, through the guidance of the sliding groove **134**, the connecting member **152** swings up and down relative to the sliding groove **134** of the cylinder, making the upward and downward

sliding of the rotating plate **151** relative to the cylinder more stable and precise.

(97) Referring to FIG. **35**, the tilt angle adjustment member **126** can swing up and down relative to the supporting arm **122**, which can drive the display mounted on the rotation adjustment assembly **15** upward or downward, and can adjust a sum of a total height of the supporting arm **122** and the lifting arm **123**. Referring to FIG. **36**, the pitch adjustment assembly **14** can rotate upwardly or downwardly relative to the second supporting rod **13**, which can drive the display mounted on the rotation adjustment assembly **15** upwardly or downwardly. In the fourth embodiment, the pitch adjustment assembly **14** is tilted upward at a maximum upward pitch angle of 45° , the pitch adjustment assembly **14** is tilted downward at a maximum downward pitch angle of 45° . The angle is an angle between a vertical direction and a center line of the connecting member **152**.

(98) Therefore, in the present disclosure, it is convenient for the user to adjust the pitch angle of the display to be able to find an appropriate angle of use more easily in the use of standing, lying down, and sitting postures. The viewing experience is more comfortable.

(99) Referring again to FIG. **29**, the mounting plate **141** defines a limit slot **1413**. The limit slot **1413** is a curved slot centered on a center of the mounting plate **141**. The mounting plate **151** defines a fixing hole **1516**. The fixing hole **1516** corresponds to the limit slot **1413**. The rotation adjustment assembly **15** further includes a limiting member **154**. The limiting member **154** passes through the limit slot **1413** and is secured to the fixing hole **1516** in the rotating plate **151**. The limiting member **154** may be located at different positions in the limit slot **1413** when the rotating plate **151** is rotated relative to the mounting plate **141**.

(100) In at least one embodiment, the limiting slots **1413** are two. The two limiting slots **1413** are symmetrically located relative to the center of the mounting plate **141**. The fixing holes **1516** are two. The two fixing holes **1516** are symmetrically located relative to the center of the rotating plate **151**.

(101) Therefore, with the two limiting slots **1413**, the two fixing holes **1516** and the two limiting members **154**, it is possible to make a rotatable connection between the limiting member **154** and the rotating plate **151**, and the connection is more stable.

(102) In at least one embodiment, the rotation angle of the rotating plate **151** is clockwise rotation 0° to 90° , which can realize a horizontal and vertical screen conversion of the display, and can make the display bring different visual effects in the horizontal or vertical screen conversions.

(103) In at least one embodiment, referring to FIG. **38**, the mobile bracket **1** further includes at least three roller sets **17**. The at least three roller sets **17** are located at different locations on a lower surface of the bottom base **11** and spaced apart from the power supply assembly **16**. The mobile bracket **1** further includes a shielding ring **112**. The shielding ring **112** is bent downwardly from a periphery of the bottom base **11**. A distance between a bottom end of the shielding ring **112** and the ground is smaller than a distance between the bottom base **11** and the ground.

(104) Therefore, a rapid movement of the mobile bracket **1** can be realized by the at least three roller sets **17**. Utilizing the shielding ring **112** to shield the at least three roller sets **17** allows the shielding ring **112** to be almost near the ground while maintaining the aesthetics of the movement.

(105) Referring to FIG. **39**, the supporting arm **122** and the lifting arm **123** define threading holes through which wires are routed between the power supply assembly **16** and the display fixed to the rotation adjustment assembly **15**. The power supply assembly **16** is configured to connect to a municipal circuit to supply power to the display mounted on the rotation adjustment assembly **15**.

(106) Therefore, the internal wiring can avoid a trouble of managing power wires, and hide the wires while also avoiding cable tangling or broken wires during the functional use of the mobile bracket **1**. At the same time, the bottom base **11** has a space reserved for batteries, which can be built-in if needed to allow the display to break free from the wire harness, anytime, anywhere, as you wish to use.

(107) Referring to FIG. **29**, FIG. **31** and FIG. **34**, the rotation adjustment assembly **15** further includes a shield **153**. The shield **153** is located on a side of the mounting plate **141** near the first

supporting rod **12** and is hollowed at a position corresponding to the connecting arm **140**. The shield **153** covers the mounting plate **141** and the rotating plate **151** on a back of the display.

(108) Therefore, through the shield **153**, the connection relationship between the mounting plate **141** and the rotating plate **151** can be obscured, making the appearance more beautiful.

(109) The above description of the technical solution of the subject matter of the present disclosure as well as the corresponding details are described above, and it can be understood that the above description is only at least one embodiment of the technical solution of the subject matter of the present disclosure, and some of the details can also be omitted in its specific implementation.

(110) In addition, in at least one embodiment of the above present disclosure, there are multiple embodiments of the combination of implementation possibilities, various combination programs are limited to space will not be listed. The technical personnel in the field can freely combine the implementation of the above embodiments according to the needs of the specific implementation, in order to obtain a better application experience.

(111) In summary, it can be understood that the present disclosure has the above mentioned excellent characteristics, so that it can be used to enhance the effectiveness of the previous technology has not been practical, and become a very practical value of the product.

(112) The above is only a better example of the present disclosure, and is not intended to limit the present disclosure. Any modification, equivalent substitution or improvement made within the ideas and principles of the present disclosure shall be included in the scope of protection of the present disclosure.

Claims

1. A mobile bracket, wherein: the mobile bracket comprises a bottom base, a first supporting rod, a second supporting rod, a pitch adjustment assembly and a rotation adjustment assembly, the bottom base is connected to a bottom end of the first supporting rod, the second supporting rod is connected to a top end of the first supporting rod, the pitch adjustment assembly is connected to one end of the second supporting rod far from the first supporting rod, the pitch adjustment assembly is capable of rotating up and down relative to the second supporting rod so as to adjust a pitch angle; the pitch adjustment assembly comprises a mounting plate located at one end far from the first supporting rod; the rotation adjustment assembly is rotatably connected to the mounting plate; the rotation adjustment assembly is capable of rotating around an axis perpendicular to the mounting plate so as to adjust a rotation angle; and the rotation adjustment assembly is configured to be connected to a back of a display; wherein: the mobile bracket further comprises a power supply assembly, the power supply assembly is located within the bottom base, the first supporting rod defines a first threading hole; the second supporting rod defines a second threading hole; the pitch adjustment assembly defines a third threading hole; the first threading hole, the second threading hole and the third threading hole are interconnected; the first threading hole, the second threading hole and the third threading hole are configured for a wire connected between the power supply assembly and the display connected to the rotation adjustment assembly passing through.
2. The mobile bracket according to claim 1, wherein: the pitch adjustment assembly comprises a rotating shaft and a connecting arm, the rotating shaft is connected to the end of the second supporting rod far from the first supporting rod, the connecting arm is rotatably connected to the rotating shaft, and the connecting arm is capable of rotating around a center axis of the rotating shaft.
3. The mobile bracket according to claim 2, wherein the connecting arm comprises the mounting plate, a first arm and a second arm extending coextensively from opposite ends of the mounting plate; the first arm and second arm are spaced apart from each other, and the rotating shaft rotatably connects the first arm and the second arm to the end of the second supporting rod far from the first supporting rod.

4. The mobile bracket according to claim 3, wherein: the pitch adjustment assembly further comprises a torsion spring and a limiting rod, the torsion spring is threaded onto the rotating shaft, the torsion spring comprises a first free end and a second free end extending in opposite directions, the first free end resists on a top wall within the second supporting rod, the limiting rod is secured between the first arm and the second arm, and is located between the rotating shaft and the mounting plate and is parallel to the rotating shaft, and the second free end is held against a lower side of the limiting rod.
5. The mobile bracket according to claim 3, wherein: the pitch adjustment assembly further comprises a first shell and a second shell, the first shell is connected to the first arm, the second shell is connected to the second arm, the first shell comprises a first hook protruding from an inner wall of the first shell, the first arm defines a first hole corresponding to a position of the first hook; when the first shell is located on an outside of the first arm, the first hook passes through the first hole and is snapped onto an inner wall of the first arm; the second shell comprises a second hook protruding from an inner wall of the second arm; the second arm defines a second hole corresponding to a position of the second hook; and when the second shell is located on an outside of the second arm, the second hook passes through the second hole and is snapped onto an inner wall of the second arm.
6. The mobile bracket according to claim 5, wherein: the first shell comprises a first snap portion at a position far from the mounting plate; the first arm comprises a first inclined portion at a position corresponding to the first snap portion; the first snap portion is snapped onto the first inclined portion; the second shell comprises a second snap portion at a position far from the mounting plate, the second arm comprises a second inclined portion at a position corresponding to the second snap portion; and the second snap portion is snapped onto the second inclined portion.
7. The mobile bracket according to claim 1, wherein: the rotation adjustment assembly comprises a rotating plate and a connecting member, the connecting member connects the mounting plate and the rotating plate, the rotating plate is capable of rotating relative to the connecting member, and the rotating plate is configured to be connected to the back of the display.
8. The mobile bracket according to claim 7, wherein: the mounting plate comprises a protrusion on a side of the mounting plate facing the rotating plate; the rotating plate defines a guiding groove corresponding to the protrusion; the guiding groove is a curved groove centered on a center of the rotating plate; the protrusion is inserted into the guiding groove to form a limit to a rotational travel of the rotation adjustment assembly, and the rotation angle of the rotation adjustment assembly ranges from 0° to 90° .
9. The mobile bracket according to claim 1, wherein: the mobile bracket further comprises a fixing member, the fixing member comprises a supporting body, a first folding portion and a second folding portion located on opposite sides of the supporting body; the first folding portion and the second folding portion are L-shaped and fixed to a lower surface of the bottom base; and a space between the supporting body, the first folding portion and the second folding portion are configured for mounting the power supply assembly.
10. The mobile bracket according to claim 1, wherein: the mobile bracket further comprises at least three roller sets, and the at least three roller sets are located at different positions on the lower surface of the bottom base and spaced from the power supply assembly.
11. The mobile bracket according to claim 10, wherein: the mobile bracket further comprises a shielding ring, the shielding ring is bent downwardly from a peripheral edge of the bottom base, and a distance between a bottom end of the shielding ring and a ground is smaller than a distance between the bottom base and the ground.
12. The mobile bracket according to claim 1, wherein: the first supporting rod comprises a supporting arm, a lifting arm and a height adjusting member; the lifting arm is inserted at one end of the supporting arm far from the bottom base, the height adjusting member is located within the supporting arm and the lifting arm; and the height adjusting member adjusts a protruding length of

the lifting arm relative to the supporting arm to adjust a total height between the supporting arm and the lifting arm.

13. The mobile bracket according to claim 12, wherein the height adjusting member is a damped gas spring.

14. The mobile bracket according to claim 12, wherein: the height adjusting member comprises a self-locking gas spring and a control button; the self-locking gas spring is located within the supporting arm and the lifting arm; the control button is connected to one end of the lifting arm far from the supporting arm and pressed against the self-locking gas spring; when the control button is pressed, a protruding height of the lifting arm relative to the supporting arm is adjusted by lifting the lifting arm upwardly or pressing the lifting arm downwardly; and when the control button is released, a height adjustment function of the self-locking gas spring is locked again.

15. The mobile bracket according to claim 1, wherein the first supporting rod comprises a supporting arm, a lifting arm and a tilting angle adjusting member; the lifting arm is tilted relative to the supporting arm, the tilt angle adjusting member is received in the lifting arm; the tilt angle adjusting member adjusts a tilt angle of the lifting arm relative to the supporting arm; and the tilt angle adjusting member causes the lifting arm to be stayed at any angle within a predetermined range of angle adjustment and is capable of providing support for suspension of the display.

16. The mobile bracket according to claim 15, wherein: the lifting arm comprises a first mounting base, a lower connecting rod, an upper tension spring and a second mounting base, the first mounting base is connected to a top end of the supporting arm, the second mounting base is connected to one end of the second supporting rod near the lifting arm, the lower connecting rod is rotatably connected to a bottom end of the first mounting base and a bottom end of the second mounting base, the upper tension spring is rotatably connected to a top end of the first mounting base and the bottom end of the second mounting base, and the second mounting base is connected to the second supporting rod.

17. The mobile bracket according to claim 16, wherein: the pitch adjustment assembly comprises a rotating shaft and a connecting arm, the rotating shaft is fixed to the end of the second supporting rod far from the first supporting rod; the connecting arm comprises the mounting plate, a first arm and a second arm formed by bending and extending in a same direction from opposite ends of the mounting plate; the first arm and the second arm are spaced apart from each other; the rotating shaft rotatably connects the first arm and the second arm to the end of the second supporting rod far from the first supporting rod; the mounting plate is in a shape of a disk; the rotation adjustment assembly comprises a rotating plate and a connecting member; the connecting member connects the mounting plate and the rotating plate; the rotating plate is capable of rotating relative to the connecting member; and the rotating plate is configured to be connected to the back of the display.

18. The mobile bracket according to claim 17, wherein: the mounting plate defines a limit slot, the limit slot is a curved slot centered on a center of the mounting plate, the rotation adjustment assembly further comprises a limiting member, one end of the limiting member is fixed to the rotating plate and the other end of the limiting member is inserted into the limiting slot.

19. The mobile bracket according to claim 17, wherein: the rotation adjustment assembly further comprises a shield, the shield is located on one side of the mounting plate near the first supporting rod and corresponding to the connecting arm, and the shield covers the mounting plate and the rotating plate on the back of the display.
